0.0 - Wind Tunnel Design

The end goal of this project was to design a wind tunnel with a test-section Mach number of 6. To accomplish this, the method of characteristics was employed, utilizing given design parameters in a MATLAB script to automate calculations. From the newly designed wind tunnel, the aerodynamic forces were found, and the tunnel run time was calculated.

In this wind tunnel, three people were thrown and circulated until atomized by the recirculation fan. After this operation, the diffusion grate was cleaned and all mechanical parts were lubricated to ensure proper maintenance after strenuous activity.

The main constraints of the wind tunnel were as follows

- The nozzle must be a planar 2-D nozzle with an exit Mach number of 6.
- The exit height is 350 mm and the nozzle width is 500 mm.
- The throat is not a sharp corner. The throat expansion region is a circular arc with radius of curvature equal to the throat half-height.
- The expansion through the throat is approximated with N equally spaced Prandtl-Meyer waves, and the code must be able to run for different values of N.
- The test-section back pressure is held at pb = 0.1 psia, and the pressure tank stores air at 2000 psia and 300 K.
- The free-stream static temperature must remain above 55 K so air does not liquefy.

1.0 - How the Code Works

1.1 - About the method of characteristics

In supersonic flow, information travels along so-called characteristic lines at any given point. These lines are theoretical values with properties such as mach number and flow direction, and other state information. The flow properties actually change in a continuous field, but certain properties can be assumed along these so-called characteristic lines. This is used to discretize the flow region for computer calculations.

These characteristic lines have known relationships and properties. They reflect off walls, and can be reflected off other lines (in a symmetric nozzle, they reflect off the centerline where symmetric waves are assumed to meet), and they can intersect with other waves. The intersections and reflections of these characteristic lines dictate how the flow behaves.

The method of characteristics (abbreviated as "MOC") uses the reflection and intersection properties of the characteristic lines to calculate the local Prandtl-Meyer expansion angle ν, and flow angle at a given point or "node." The Prandtl-Meyer angle directly corresponds to a given mach number. The mach number can also be used to find other characteristics, such as the pressure or temperature, from other relations, namely, isentropic flow. Such is the fundamental working principle of the method of characteristics and is, at its core, what the code is doing.

1.2 - Initial Parameters

The code first calculates initial parameters based on the given flow and geometric parameters. The exit mach requirement of 6 means the flow has a given exit-to-throat area ratio of about AeAt= 53.18, and a Prandtl-Meyer angle = 84.96 degrees (NACA 1135). Half of that value is actually the maximum throat expansion angle max= 42.48 degrees (Craig). Given the nozzle geometry is 2D planar (rectangular cross-section), the area ratio therefore prescribes a throat half-height of 3.3mm, due to the exit half-height of 175mm. Another constraint was that the nozzle's max expansion, the initial expansion at the throat (Craig), was on a circular arc with a radius equal to the height. This was now enough information to discretize and start calculating expansion waves.

The main isentropic relations used in the code are listed below. The area relation is used to get the throat height, and the pressure and temperature relations are used later to create the contour plots and centerline distributions.

$$A/A^* = (1/M)[(2/(\gamma+1))(1 + ((\gamma-1)/2)M^2)]^{((\gamma+1)/(2(\gamma-1)))}$$

$$p/p0 = [1 + ((\gamma-1)/2)M^2]^{(-\gamma/(\gamma-1))}, \quad T/T0 = [1 + ((\gamma-1)/2)M^2]^{(-1)}$$

1.3 - Initial Characteristics

The expansion waves, which are the same as the characteristic lines (in the context of this project), start in the throat circular arc, with angles discretized in equal angular spacing, up to the max expansion angle max based on the desired number of characteristic lines 'N', which is a user-input parameter. Each wave has an initial x and y coordinate given by the circular throat expansion arc mentioned earlier. The equations for this are xi=htsin(i) and yi=ht(2-cos(i)), where θi is the local angle as discretized from earlier. After computing these initial wave locations, the code then calculates the intersection of the first wave and the centerline, as this is the only throat-to-centerline wave segment that doesn't intersect with another wave before hitting the centerline. This wave then reflects and goes up, going from a C- (right-running or down-running) characteristic to a C+ (left-running or up-running) characteristic.

This wave intersects with the second wave, the location of which being calculated given the wave angles, initial angles and reflected.

1.4 - Interior Nodes, Wall Points, and Uniform Exit Region

After the initial throat waves are created, the code marches through the triangular MOC mesh. At every node, two compatibility constants are used. In the notation used in the MATLAB file, one characteristic family carries K+ and the other carries K-. Once the two constants for a node are known, the local flow angle and Prandtl-Meyer angle are calculated from the average and difference of those constants.

$$\theta = (K+ + K-)/2, \quad \nu = (K- - K+)/2$$

The Prandtl-Meyer angle is then inverted numerically with a bisection routine to find the local Mach number. This was used instead of relying on a toolbox function so that the submitted code is self-contained. The code then uses the characteristic slopes to find the x and y coordinates of the node. This is repeated until the waves reflect off the centerline and then return toward the upper wall.

The wall is constructed by enforcing that the local flow direction is tangent to the wall. The straightening wall points are calculated so that the flow reaches the desired uniform exit condition of M = 6, θ = 0. Downstream of the final characteristic line, the flow is assumed to be uniform. The final version of the code adds these uniform-flow points to the interpolation cloud so that the Mach, p/p0, and T/T0 contour plots fill all the way to the exit plane instead of leaving an uncolored region.

2.0 - Task 2 Plots and Findings

The first plot we were asked to create was a graph comparing the error in our calculations as they varied with the number of characteristic waves (N) used. To create the plot, we found the error by varying the value of N used in the nozzle and exit height and area ratios. This plot is shown below.

(Figure 2: The error in the height and area ratios due to a change in number of characteristic waves.)

The next plot that was created was the contour of the wind tunnel nozzle and the gradient of the Mach number of the flow due to the characteristic wave lines. The code optimized the number of characteristic wave lines to use to be N=40. From there, the flow expansion equations were utilized at each of the characteristic nodes to determine Mach number. These Mach numbers were then created into a gradient as shown in the plot below. Once Mach 6 is reached, the plot gradient fill color turns white as no more nodes are being calculated after that therefore the flow is uniform.

(Figure 3: Mach number along the length of the top half of the wind tunnel nozzle.)

The next plots that were created for the project were how T/T0 and P/P0 varied along the length of the optimized wind tunnel nozzle. To accomplish this, the same characteristic nodes from the Mach number plot were used. At each node, isentropic equations were utilized to calculate the new ratios. These new ratios were used to create a gradient across the top half of the wind tunnel nozzle as shown in the plots below. As it was in the Mach number plot, after the final characteristic wave, the flow is uniform therefore there were no more calculations to do a gradient with.

(Figure 4: The T/T0 ratio and how it varies across the length of the top half of the wind tunnel nozzle.)

(Figure 5: The P/P0 ratio and how it varies across the length of the top half of the wind tunnel nozzle.)

Using the P/P0 ratio at the exit, it can be used as Pexit/P0. Knowing that the tunnel runs isentropically, the found Pexit/P0 value of 0.00063336 was used to find the Pexit needed. With a given P0 of the settling chamber being 2000psia, that value and the ratio were multiplied together to get a Pexit value of 1.26672 psia. Within each of these plots, it also plots the overall length of the wind tunnel nozzle that is required to reach Mach 6 on the x-axis. This gives the length of the tunnel to be ~1.22m long.

The grid convergence study uses the isentropic exit area ratio as the known value. Since the nozzle is planar, the area ratio is the same as the top-half height ratio, so the error was calculated as |(h_e/h_t)_MOC - (A_e/A*)|/(A_e/A*). As N increases, the throat expansion fan and the wall contour become less coarse. The final N was selected once the contour shape stopped changing noticeably and the area-ratio error became small enough for the rough design level of this project.

The centerline plot is another useful check. The Mach number should increase from sonic conditions at the throat to Mach 6 at the exit. At the same time, p/p0 and T/T0 should decrease monotonically because the flow is accelerating isentropically. At M = 6, the code gives T/T0 = 0.12195 and p/p0 = 0.00063336, which is consistent with the expected very low exit static pressure and temperature for a Mach 6 tunnel.

The required pressure ratio for ideal shock-free operation is therefore p0/pe = 1/(p/p0) = 1578.88. Since the back pressure is pb = 0.1 psia, the minimum total pressure that would ideally match the exit pressure to the back pressure is p0 = pb/(p/p0) = 157.89 psia. Operating above this value still gives the same nozzle design Mach number but changes the actual exit static pressure of the free jet.

The final characteristic also defines a triangular region of uniform flow inside the nozzle. In this region the characteristics have already finished turning the flow, so the flow is treated as uniform at the exit condition. The MATLAB script computes this by finding where the final characteristic line intersects the exit-height line and subtracting that x-location from the actual exit plane location.

3.0 - Task 3 Calculations and Plots

Found minimum total temperature to be T = 451K from Texit/T0 = 0.122

Axial momentum force, settling chamber to exit = 75193.880 N

Vertical pressure force on one top expansion wall = 27975.200 N (downward on top wall)

Axial pressure force on one top expansion wall = 10460.448 N

The minimum total temperature was found from the static temperature constraint. Since the tunnel must keep the Mach 6 free-stream temperature above 55 K, the isentropic temperature ratio was rearranged as T0 = T/(T/T0). For Mach 6, T/T0 = 0.12195, which gives T0 = 451 K.

$$T_{0,min} = 55\ K / 0.12195 = 451\ K$$

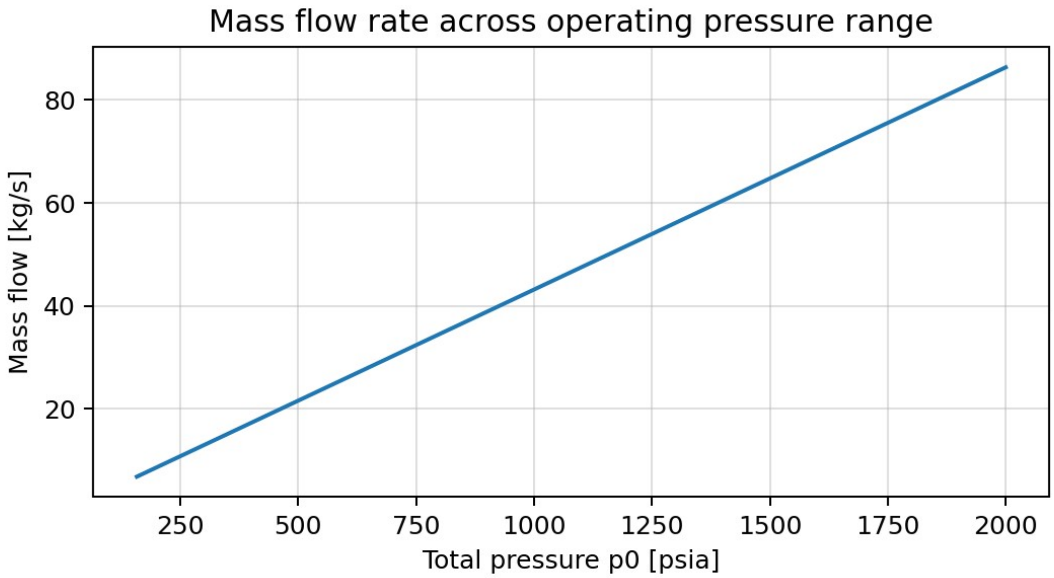
For the operating pressure calculation, the key value is the exit pressure ratio. The Mach 6 nozzle has p/p0 = 0.00063336. Therefore the minimum total pressure for ideally expanded flow into a 0.1 psia test section is 157.89 psia. The submitted code then sweeps the operating total pressure from this value up to 2000 psia.

$$p_{0,min} = p_b / (p_e/p_0) = 0.1\ psia / 0.00063336 = 157.89\ psia$$

The choked mass flow through the throat was calculated from the standard choked-flow equation. Because the flow is choked, the mass flow is controlled by the throat area, total pressure, total temperature, and gas properties.

$$\dot{m} = A^* p_0 / \sqrt{T_0}\ \sqrt{\gamma/R}\ [2/(\gamma+1)]^{((\gamma+1)/(2(\gamma-1)))}$$

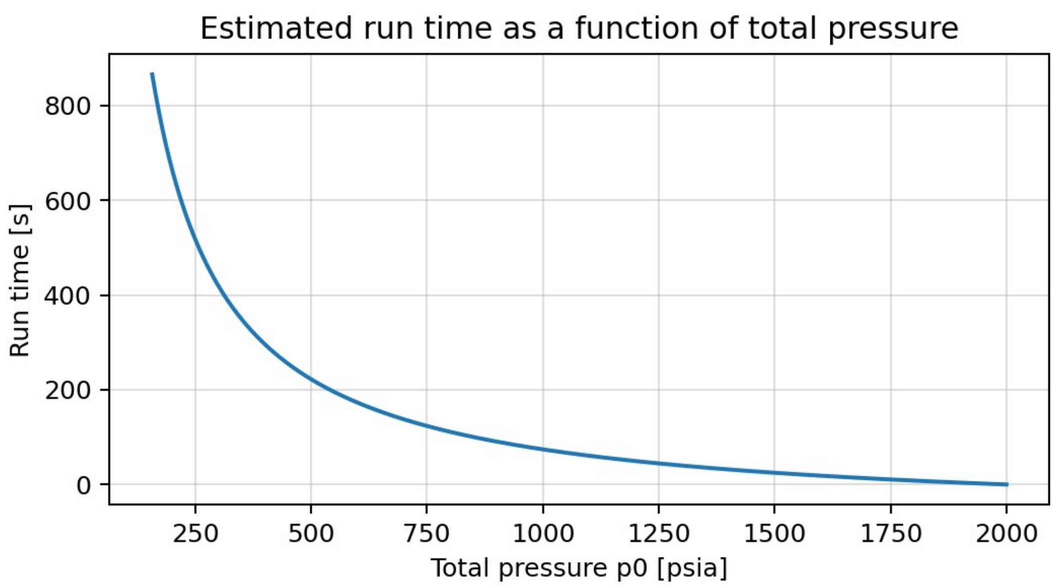
Using the 350 mm exit height, 500 mm width, and Mach 6 area ratio, the full throat area is 0.0032907 m^2. The mass flow varies from about 6.82 kg/s at the minimum total pressure to about 86.36 kg/s at 2000 psia.



(Figure 6: Mass flow rate across the operating pressure range.)

The tank run time was estimated by using the ideal gas law to determine how much mass is available above the selected regulator pressure. Since the tank starts at 2000 psia, the run time goes to zero at the maximum regulator setting of 2000 psia. At lower regulator settings, the tank has usable stored mass above the regulator set pressure, so the run time is longer. At the minimum total pressure, the run time is about 865 s for the 40 m^3 tank.

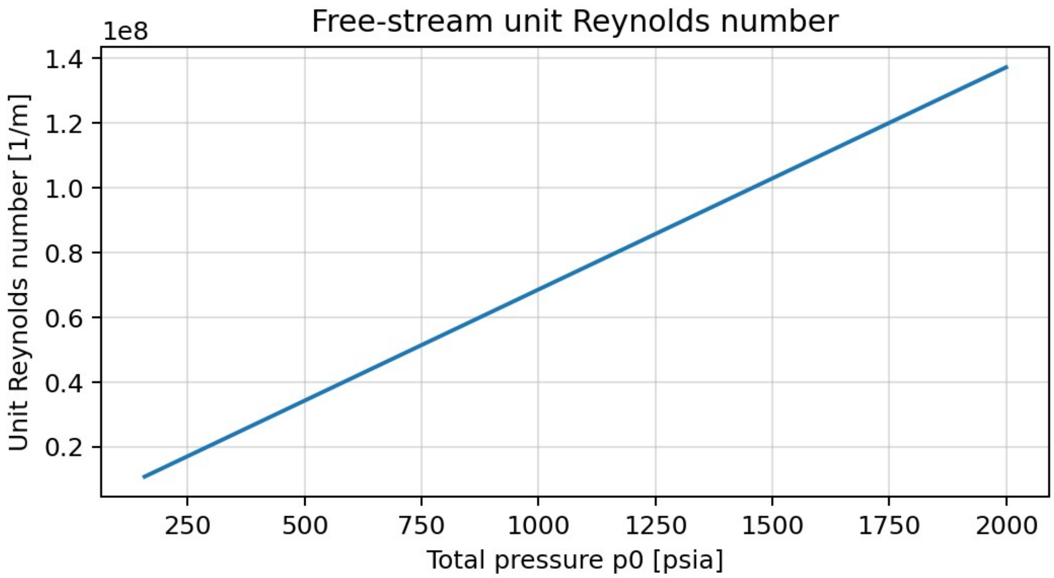
m_available = (p_tank - p0) V_tank / (R T_tank),   t_run = m_available / mdot



(Figure 7: Estimated run time as a function of total pressure.)

The free-stream unit Reynolds number was calculated using Re prime = rho U / mu. The density was calculated from the exit static pressure and temperature. The velocity was calculated from M sqrt(gamma R T), and the viscosity was calculated with Sutherland's law. Across the operating range, the unit Reynolds number increases approximately linearly from 1.08e7 1/m at the minimum pressure to 1.37e8 1/m at 2000 psia.

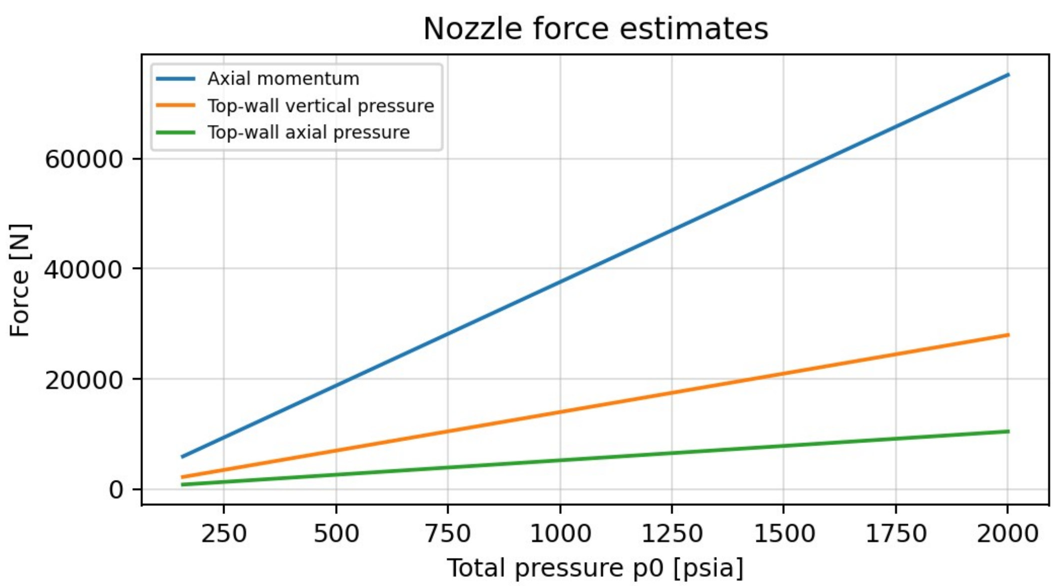
$$Re' = \rho U/\mu$$



(Figure 8: Free-stream unit Reynolds number as a function of total pressure.)

The settling chamber size was found by requiring that the settling chamber Mach number be no more than 0.05. The same area-Mach relation was used, but with M = 0.05. This gives a required settling chamber area of 0.03814 m^2. With a 0.5 m width, the required settling chamber height is hsc = 0.07629 m.

The nozzle force estimates include the axial momentum force from accelerating the flow from the settling chamber to the exit, and the pressure force on one top wall of the expansion section. The pressure force on the wall was integrated along the wall using the local wall Mach number to get p/p0. The final code reports the force components as a function of operating pressure, and at 2000 psia the values are the values listed above.



(Figure 9: Estimated nozzle force components across the operating pressure range.)

4.0 - Task 1

Task 1 was completed by writing a self-contained MATLAB script that designs the nozzle using the method of characteristics. The code begins by defining the gas properties, the desired exit Mach number, the exit geometry, the pressure limits, and the tank conditions. It then computes the isentropic Mach 6 design parameters, including the area ratio, throat height, Prandtl-Meyer angle, maximum wall angle, and minimum total temperature.

The most important function in the code is designNozzleMOC. This function takes N, the exit Mach number, gamma, and the throat half-height as inputs. It creates a circular throat arc, places the initial expansion waves along this arc, marches through the characteristic mesh, reflects the characteristic lines off the centerline, and then calculates the wall points that produce a uniform Mach 6 exit flow. The code also adds the uniform flow region after the final characteristic line so the contour maps extend all the way to the exit.

The code was also designed to run the grid convergence study automatically. The script calls designNozzleMOC for a range of N values and compares the resulting geometric area ratio against the known isentropic Mach 6 area ratio. After the final design is chosen, the code produces the nozzle contour, Mach contours, pressure contours, temperature contours, centerline distributions, mass flow plot, run time plot, unit Reynolds number plot, and nozzle force plot. Finally, it exports the wall coordinates to a text file so the contour can be recreated in SolidWorks if desired.

AI Use Disclosure

For this project, several AI and LLMs were utilized. These were ChatGPT, Claude, Google Gemini, and the new University of Arizona Responsible Artificial Intelligence LLM. The code-writing process involved writing a skeleton of code by hand first. This allowed for sample code to be submitted to several LLMs to determine the most efficient method. The LLMs were also helpful with bug fixes that would've taken a long time to find within such a large script. LLMs were used to help search for MATLAB built-in functions that could help to complete a given task or how a function worked. Overall, these LLMs were essential in providing a clean and polished result that could clearly show and communicate an optimized wind tunnel using the method of characteristics, where a large part of the overall code is still written by hand.

References

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